

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination May 2017

PH323 Medicinal Chemistry-II

Date: 08.05.2017 ,Monday

Time:10.00a.m. to1.00p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION - I

Q 1 Attempt all questions. Each question is of one mark. 20

1. How many fluorine atoms are in enflurane ?
[A] 4
[B] 3
[C] 2
[D] 5
2. Under which therapeutic category Thiopental fall from the followings:
[A] General anaesthesia
[B] Local anaesthesia
[C] Anticonvulsant
[D] Antiinflammatory
3. Benzocaine is chemically,
[A] Ester
[B] Amide
[C] Anilide
[D] Acid
4. Which starting material is use for preparation of procaine ?
[A] p- amino benzaldehyde
[B] p-amino benzoate
[C] o- amino benzaldehyde
[D] p-Nitro benzoic acid
5. Barbitol is,
[A] Long acting
[B] Intermediate acting
[C] Short acting
[D] Ultra short acting
6. Which heterocyclic nucleus is present in Flurazepam ?
[A] Benzodiazepam
[B] Benzoxepam
[C] Oxetane
[D] Benzothieapam

7. Which heterocyclic nucleus present in Zopiclone ?
 [A] PyrroloPyrazine
 [B] Quinoline
 [C] Benzodiazepam
 [D] Pyrimidine
8. Which of followings is class of Hydroxyzine,
 [A] Antihistaminic
 [B] Benzodizepam
 [C] Anticonvulsant
 [D] Local anaesthetic
9. Substitution of Electron withdrawing group in A ring of benzodizepam has effect on,
 [A] Increase activity
 [B] Decrease activity
 [C] Decrease solubility
 [D] Increase metabolism
10. Which functional group is present in 5th position of phenytoin ?
 [A] Phenyl
 [B] Ethyl
 [C] Allyl
 [D] Methyl
11. Trimethadione is chemically,
 [A] Quinoline
 [B] Pyrimidine
 [C] Pyrrole
 [D] Oxazolidinedione
12. Substitution of lower alkyl group at 5th position will make hydantoin to be effective on,
 [A] Petitmal epilepsy
 [B] Grandmal epilepsy
 [C] Both
 [D] Less active on both
13. Esterification of third hydroxy group in morphine will make molecule,
 [A] More active than morphine
 [B] Less active than morphine
 [C] Moderate active than morphine
 [D] Unstable metabolically than morphine
14. Which heterocyclic nucleus present in meperidine ?
 [A] Pyridine
 [B] Piperidine
 [C] Morpholine
 [D] Pyrimidine
15. What is starting material for synthesis of Atenolol ?
 [A] Hydroxy phenyl glycine
 [B] Butyramidophenol
 [C] hydroxyethyl phenol
 [D] Formyl benzoic acid
16. Indometacin is chemically in nature,
 [A] Indandione
 [B] Indole
 [C] Benzimidazole
 [D] Pyrrole
17. Warfarin is,
 [A] Antihypertensive agent
 [B] Antiplatelet agent
 [C] Anticoagulant
 [D] None

18. Which functional group is present on 6th position of quinidine ?
 [A] Methyl
 [B]Hydroxy
 [C]Chloro
 [D] None
19. Larger alkyl group in side chain of dibenzazepine will,
 [A] Increase activity
 [B] Decrease activity
 [C] Change in duration of action
 [D] Becomes toxic
20. Which side chain is present in Glibenclamide ?
 [A]Sulfonylureas
 [B]Acetamide
 [C]Propylamine
 [D]Thionylurea

SECTION – II

- Q 2 Attempt SAR of any TWO of the following**
- | | | |
|---|---------------------------------------|----|
| A | Benzodiazepines | 05 |
| B | Local anesthetic | 05 |
| C | Phenothiazine as antipsychotic agents | 05 |
- Q 3 Classify any TWO of the following**
- | | | |
|---|-------------------------|----|
| A | Antihypertensive agents | 05 |
| B | Opioid analgesic | 05 |
| C | NSAIDs | 05 |

SECTION – III

- Q 4 Attempt any FOUR of the following**
- | | | |
|---|--|----|
| A | Write down synthetic scheme for sotalol and diltiazem. | 05 |
| B | Write mechanism of action and Use of anticoagulants. | 05 |
| C | Add a short note on antiparkinson drugs. | 05 |
| D | Add a short note on anticonvulsants. | 05 |
| E | Write down SAR of glucocorticoids. | 05 |
| F | Write down chemistry and uses of CNS Stimulants. | 05 |
- Q 5 Attempt any FOUR of the following**
- | | | |
|---|--|----|
| A | Write SAR of Tricyclic antidepressants. | 05 |
| B | Write synthetic scheme for any two NSAIDs. | 05 |
| C | Add a short note on antirheumatic drugs. | 05 |
| D | Write down chemistry of Cox-2 inhibitors. | 05 |
| E | Write down chemistry of Cardiotonics. | 05 |
| F | Explain chemical nomenclature of steroid. | 05 |